# CLIENT HANDBOOK

## Smart Attendance Face Recognition System

### 1.0 Introduction

The Smart Attendance Face Recognition System is designed to automate attendance tracking using facial recognition technology. It allows users to manage classes, record attendance efficiently, and generate reports, while administrators can monitor and manage the overall system.

### 2.0 User Roles

There are two main types of users:

- **User (Lecturer / Admin)** – Manages attendance, groups, and reports
- **Super Admin** – Manages system settings, users, and overall system control

### 3.0 User Guide (Lecturer / Admin)

#### 3.1 Login & Registration

Users can access the system through:

- Email and password login
- Google login
- Microsoft login

**Features:**

- Error messages for incorrect password or unregistered email
- Validation for missing input fields
- Sign-up requires acceptance of terms and conditions

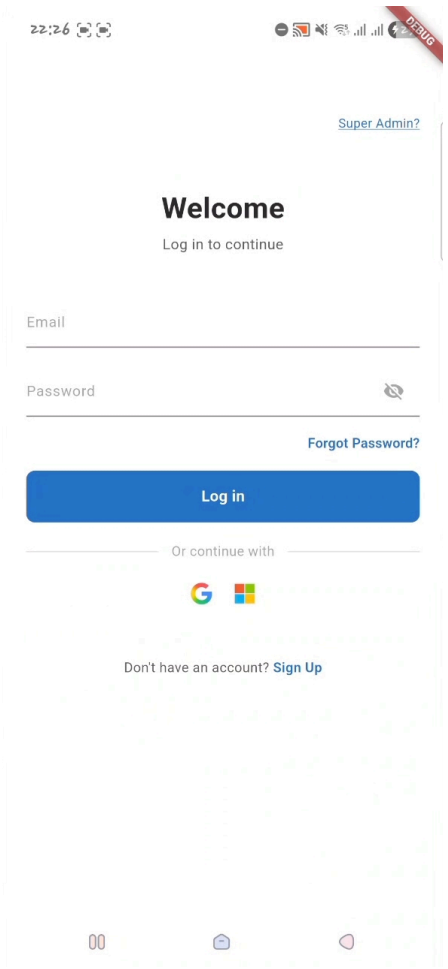


Figure 1.0: Login Page

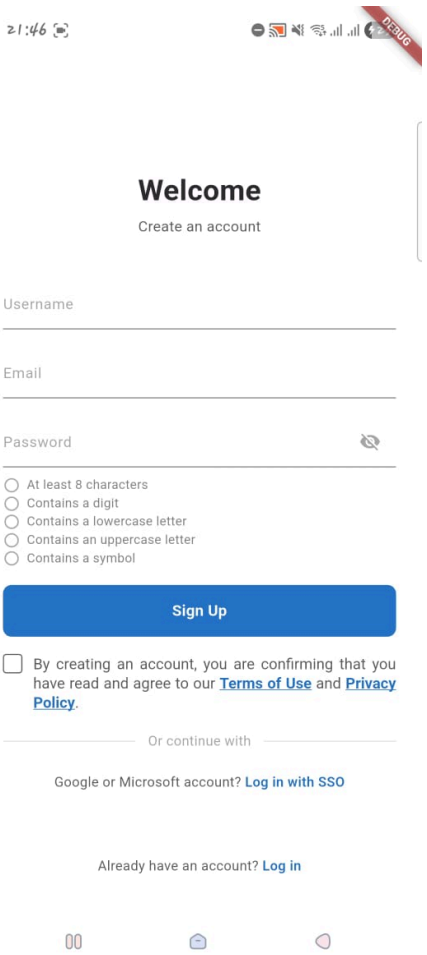


Figure 1.1: Sign Up Page

### 3.2 Dashboard

The dashboard is the main interface for managing groups and schedules.

**Key Functions:**

- Search for classes/groups
- Create new groups
- Enroll members into groups
- Edit group details (name and image)
- Delete groups with confirmation prompt
- Create and manage schedule times

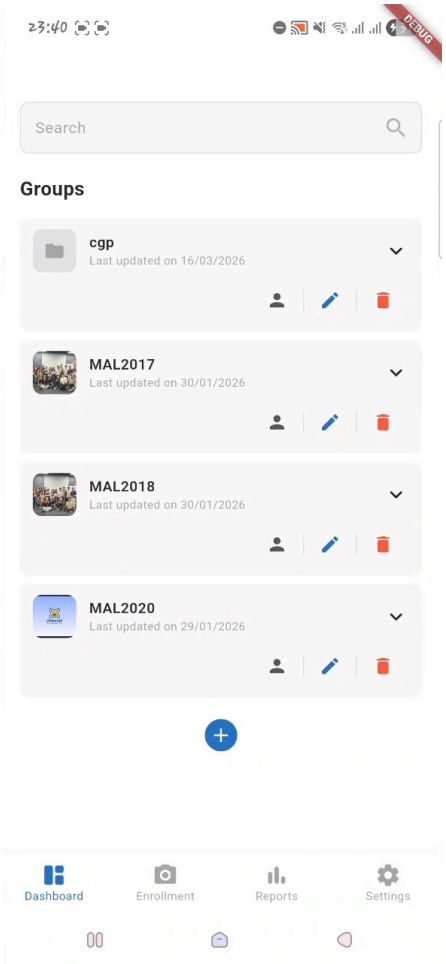


Figure 2.0: Dashboard Page

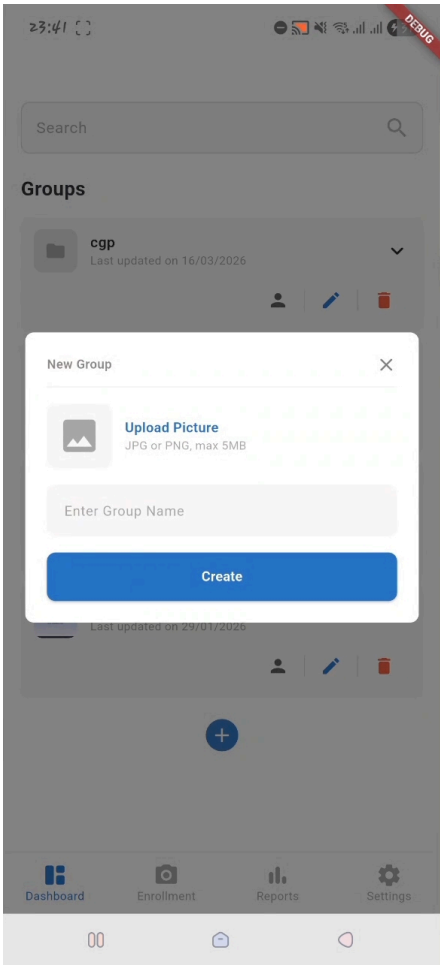


Figure 2.1: Group (Subject) Creation

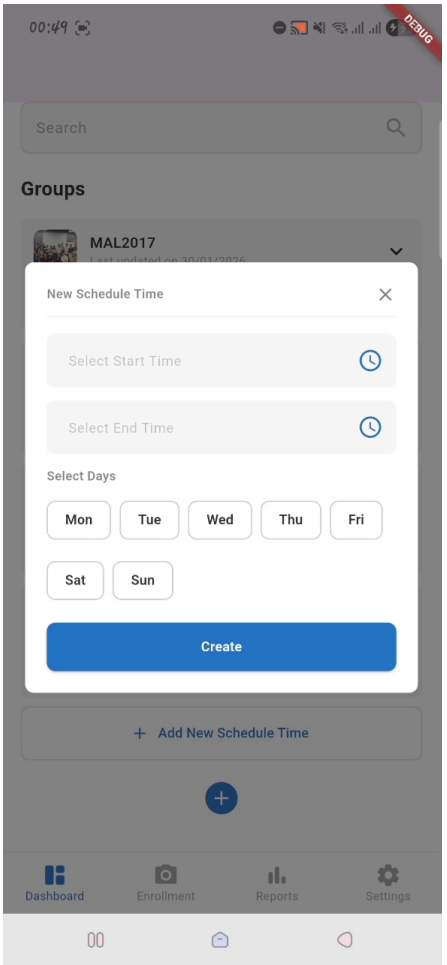


Figure 2.2: Class Schedule Creation

### 3.3 Schedule & Attendance

This module allows users to manage attendance sessions.

**Features:**

- Search and view group members
- View attendance status (Present / Absent)
- Filter and sort members (e.g., A–Z, Present/Absent)
- Manually update attendance (long-press function)
- Delete members (with confirmation)

**Taking Attendance:**

- Use the device camera to scan and recognise faces
- Supports:
    - Flashlight
    - Camera switching
    - Upload images from device
- Allows **multiple images** for better recognition accuracy
- Image confirmation and retake options

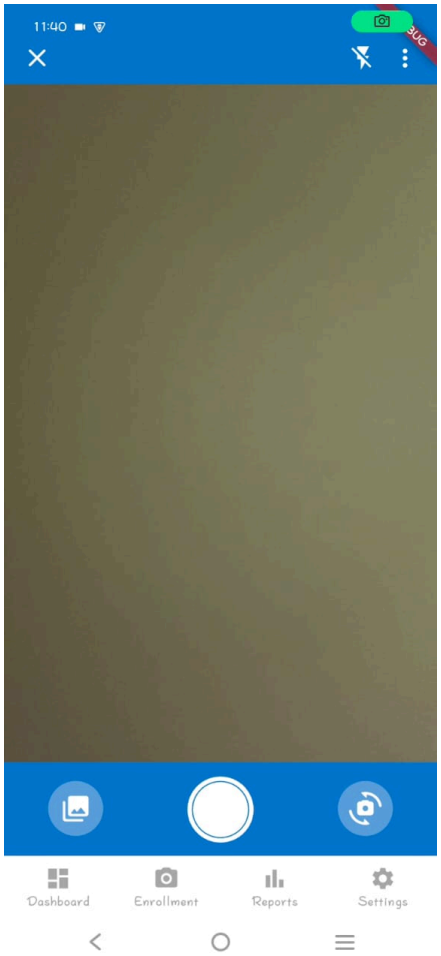


Figure 3.0: Camera Capture Page

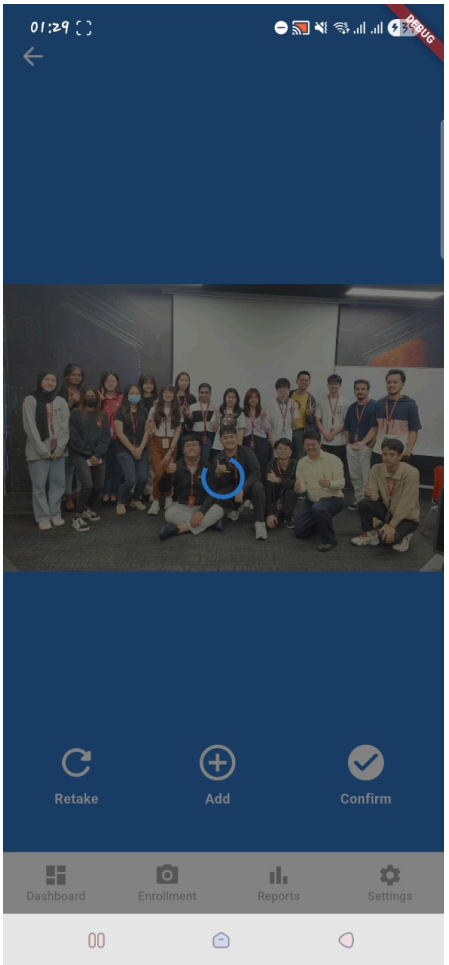


Figure 3.1: Face Scanning Page

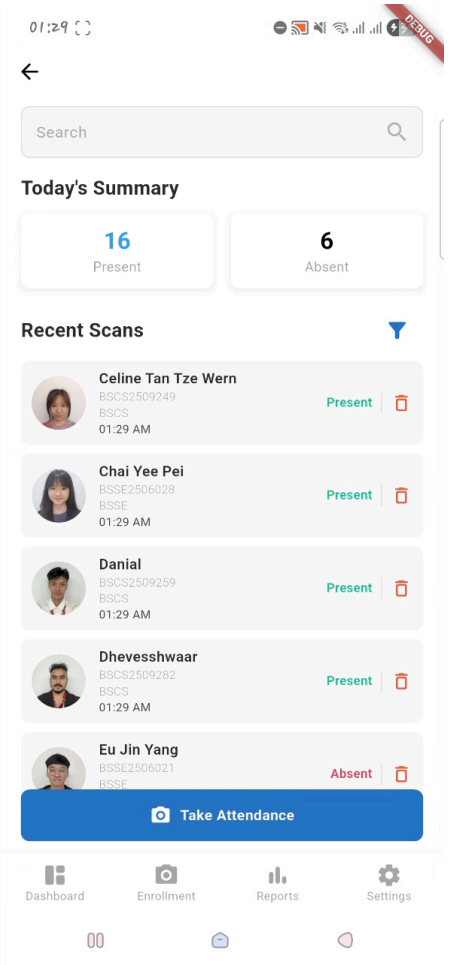


Figure 3.2: Attendance List Page

### 3.4 Enrolment

Used to register new members into the system.

**Features:**

- Capture or upload multiple images for each user
- Confirm or retake images
- Enter member details
- Save enrolment data

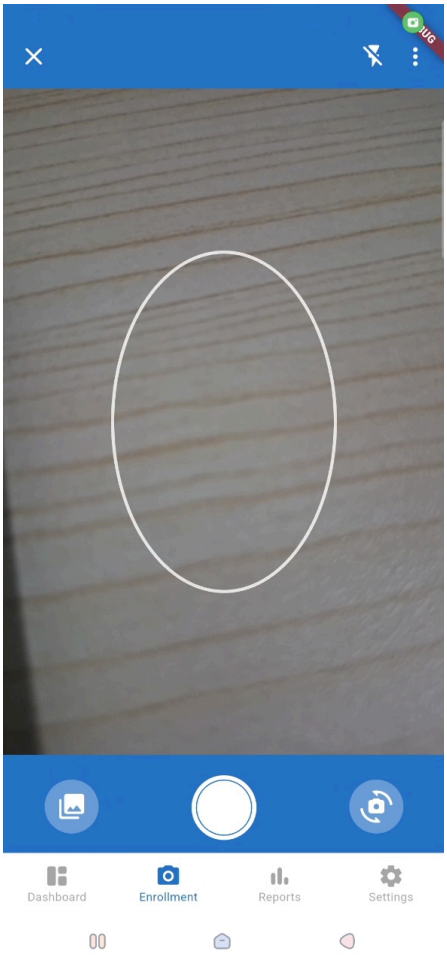


Figure 4.0: Camera Capture Page

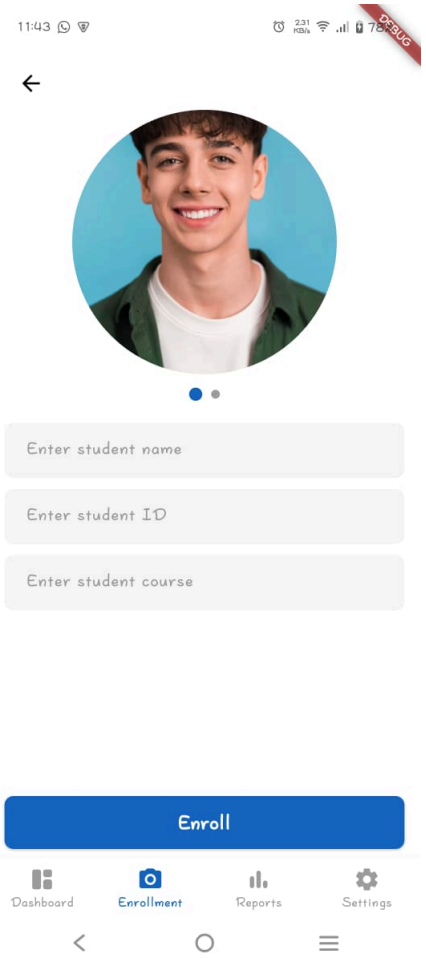


Figure 4.1:Student Form (Multiple Images)

### 3.5 Reports

This module provides attendance analytics and reporting.

**Features:**

- Select a date using the calendar
- Filter by group and schedule
- View:
    - Total attendance
    - Daily attendance rate (%)
    - Overall average performance
- Visual statistics (line graph)
- Export and download reports as PDF

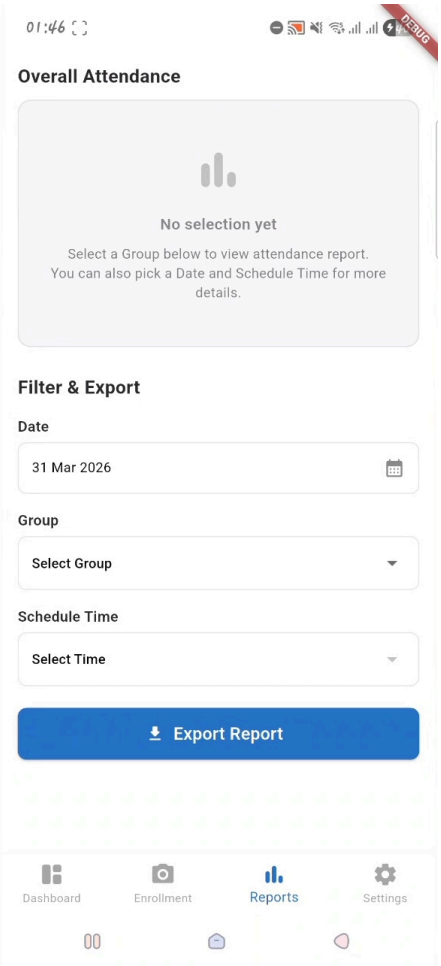


Figure 5.0: Report Page

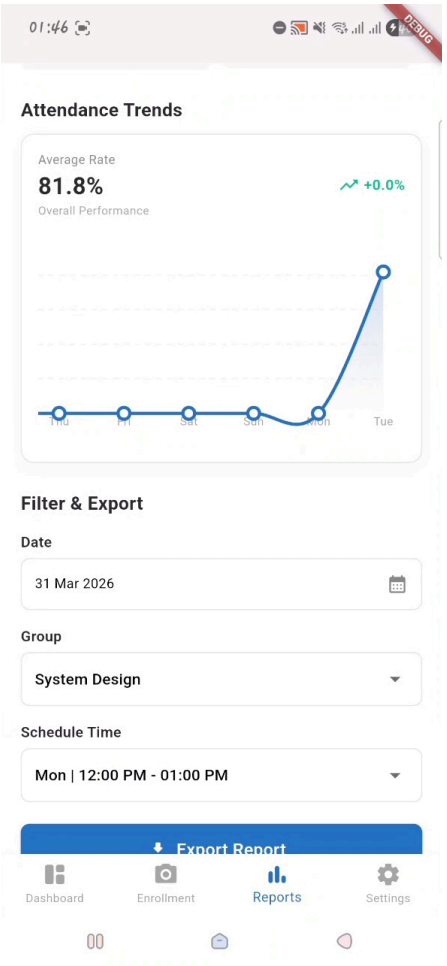


Figure 5.1: Generated Graph/Chart

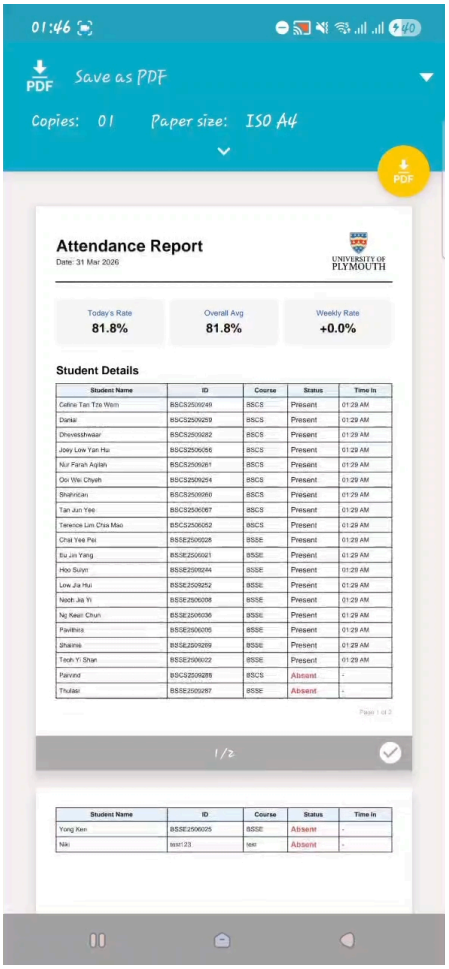


Figure 5.2: PDF Export Page

### 3.6 Settings

Users can manage personal account settings.

**Features:**

- View and update profile photo
- View username and email
- Change password
- Enable/disable attendance reminders
- Sign out

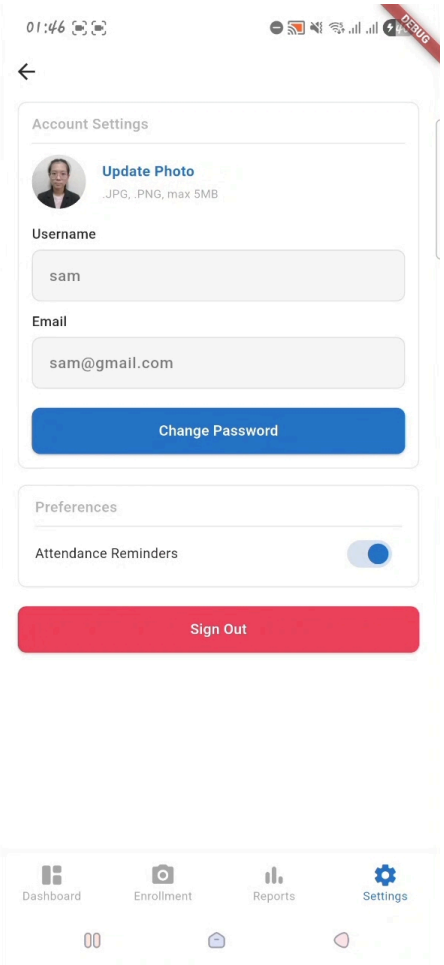


Figure 6.0: Settings Page

## 4.0 Super Admin Guide

### 4.1 Login

Same login functionality as users with validation for:

- Incorrect credentials
- Missing input
- Unregistered accounts

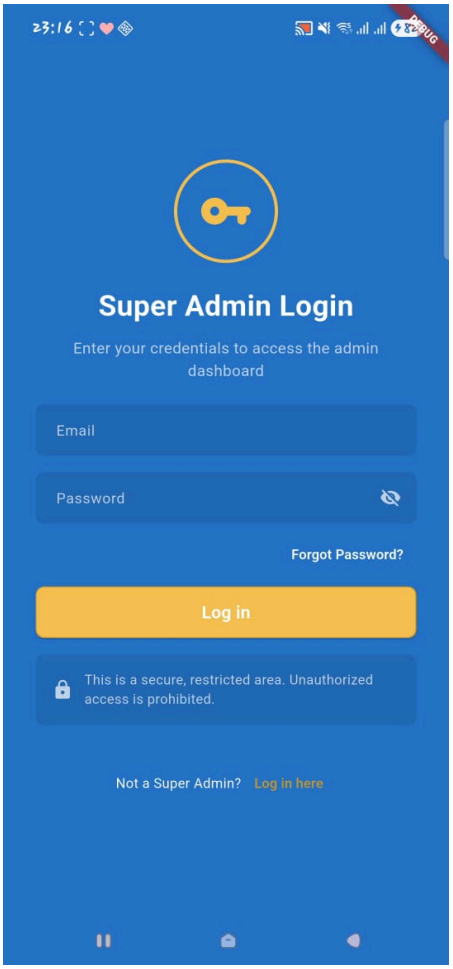


Figure 7.0: Super Admin Login Page

### 4.2 Dashboard

Provides an overview of system statistics.

**Features:**

- View profile details
- Change password
- View system statistics:
    - Total users
    - Total admins
    - Total enrolments
- Add new admin users

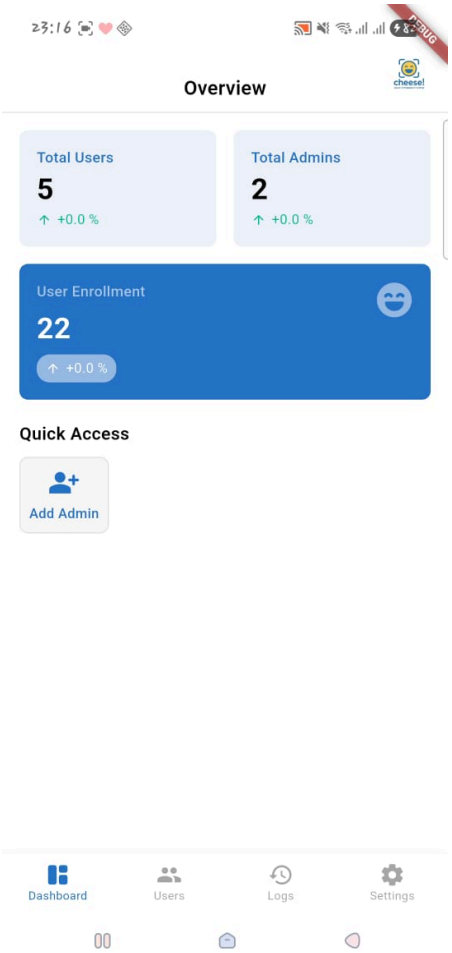


Figure 8.0: Super Admin Dashboard Page

### 4.3 User Management

Manage all system users.

**Features:**

- Search users
- Sort and filter:
    - Role (Admin/User)
    - Login type (Email/Google/Microsoft)
- Edit user details
- Delete users (with confirmation)

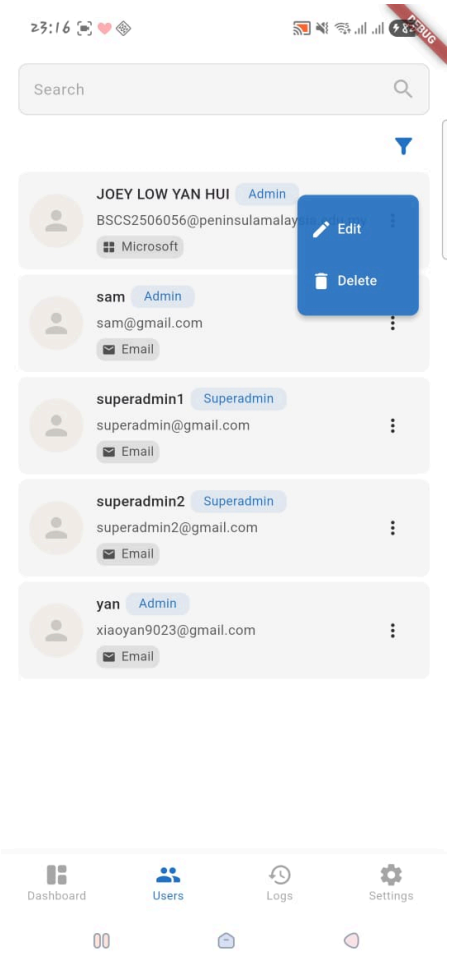


Figure 9.0: Super Admin User Management Page

### 4.4 Logs Management

Monitor system activity.

**Features:**

- Search logs
- Sort by date (latest or earliest)

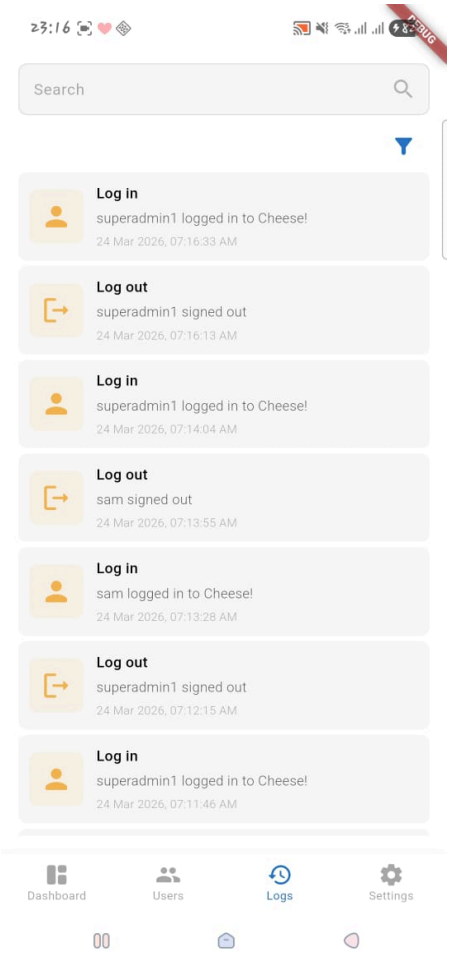


Figure 10.0: Super Admin Logs Page

### 4.5 System Settings

Configure system-level settings.

**Features:**

- Enable/disable Google and Microsoft login
- Set log retention period
- Enable automatic log deletion
- Manual log purge (with confirmation)

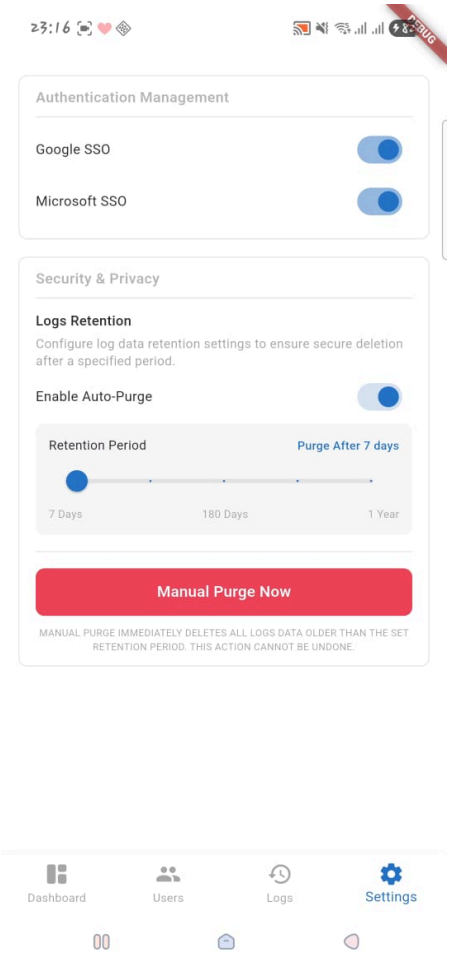


Figure 11.0: Super Admin Settings Page

## 5.0 Key Features of the System

- Automated attendance using face recognition
- Support for multiple images per user to improve accuracy
- Secure login (Email, Google, Microsoft)
- Real-time attendance tracking
- Data reporting and export (PDF)
- User and system management tools

## 6.0 Recommendations for Use

- Ensure good lighting conditions during face scanning for better accuracy
- Capture multiple images per user during enrolment to improve recognition performance
- Regularly update user data and schedules to maintain accuracy
- Review attendance reports to monitor trends and performance

## 7.0 System Requirements

To ensure the Smart Attendance Face Recognition System functions effectively, the following system requirements are recommended:

### 7.1 Hardware Requirements

- Smartphone or device with a camera (for face recognition and enrolment)
- Computer or laptop (for system access and management)
- Minimum 4GB RAM (8GB recommended for better performance)
- Stable internet connection

### 7.2 Software Requirements

- Operating System: Android / iOS (mobile application), Windows / macOS (admin access)
- Web browser (e.g., Chrome, Edge, Safari)
- Internet access for API communication and authentication services

### 7.3 System Requirements

- Active internet connection for real-time face recognition and data synchronisation
- Enabled camera access for attendance and enrolment
- Valid user account for login (Email, Google, or Microsoft)

### 7.4 Optional Requirements

- Good lighting environment for accurate face recognition
- Updated device software for better compatibility and security